

Period-3 divisibility of the S_2 -isotypic multiplicities of $R_{(k,k)}$

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Abstract

Let $R_{(k,k)} = R_{2k}^{S_k \times S_k}$ denote the parabolic coinvariant algebra of S_{2k} at the composition $\alpha = (k, k)$, viewed as a graded S_2 -module via the block-swap action. Let $m_{(2)}(t), m_{(1,1)}(t) \in \mathbb{Z}[t]$ be the trivial and sign isotypic Hilbert polynomials. We prove

$$[3]_t \mid m_{(2)}(t) \quad \text{and} \quad [3]_t \mid m_{(1,1)}(t) \quad \Longleftrightarrow \quad k \equiv 2 \pmod{3}.$$

The proof reduces both $m_\pi(t)$ to alternating partial sums of q -binomial coefficients via the Kirillov–Reshetikhin two-row fake degree, evaluates at ζ_3 via the “correct” q -analogue of Lucas’ theorem, and observes that at $k \equiv 2 \pmod{3}$ every full block of three consecutive i ’s contributes zero, whereas at $k \equiv 0, 1 \pmod{3}$ a nonzero residue survives.

1 Setup and statement

Fix $k \geq 1$ and write $n = 2k$, $\alpha = (k, k)$. Let $R_n = \mathbb{C}[x_1, \dots, x_n] / \langle e_1, \dots, e_n \rangle$ be the coinvariant algebra of S_n , and $R_\alpha := R_n^{S_\alpha}$ the $S_\alpha = S_k \times S_k$ invariants. The normaliser $N_{S_n}(S_\alpha) = S_k \wr S_2$ acts on R_α ; the residual $S_2 = N/S_\alpha$ acts by block-swap.

Theorem 1 ([1], Theorem D at $r = 2$). *The graded S_2 -isotypic Hilbert polynomials of R_α are*

$$\begin{aligned} m_{(2)}(t) &= \sum_{i=0}^{\lfloor k/2 \rfloor} \tilde{f}_{(2k-2i, 2i)}(t), \\ m_{(1,1)}(t) &= \sum_{i=0}^{\lfloor (k-1)/2 \rfloor} \tilde{f}_{(2k-2i-1, 2i+1)}(t), \end{aligned}$$

where $\tilde{f}_\lambda(t) = t^{n(\lambda)} [n]_t! / \prod_{c \in \lambda} [h(c)]_t$ is the fake degree of λ .

Our result:

Theorem 2 (Period-3 divisibility, $r = 2$). *For every $k \geq 1$,*

$$[3]_t \mid m_{(2)}(t) \quad \text{and} \quad [3]_t \mid m_{(1,1)}(t) \quad \Longleftrightarrow \quad k \equiv 2 \pmod{3}.$$

2 The two-row fake degree

Lemma 3. For $a \geq b \geq 0$ integers, letting $\begin{bmatrix} n \\ k \end{bmatrix}_t = [n]_t! / ([k]_t! [n-k]_t!)$ denote the Gaussian binomial (with $\begin{bmatrix} n \\ -1 \end{bmatrix}_t := 0$),

$$\tilde{f}_{(a,b)}(t) = \begin{bmatrix} a+b \\ b \end{bmatrix}_t - \begin{bmatrix} a+b \\ b-1 \end{bmatrix}_t.$$

Proof. For $\lambda = (a, b)$, $n(\lambda) = b$, and the hook lengths are $\{1, 2, \dots, a+1\} \setminus \{a-b+1\}$ in the first row and $\{1, 2, \dots, b\}$ in the second row. Thus

$$\tilde{f}_{(a,b)}(t) = t^b \cdot \frac{[a+b]_t!}{[b]_t! \cdot [a+1]_t! / [a-b+1]_t!} = \frac{t^b \cdot [a-b+1]_t \cdot [a+b]_t!}{[b]_t! \cdot [a+1]_t!}.$$

On the other hand, from $\begin{bmatrix} a+b \\ b \end{bmatrix}_t = [a+b]_t! / ([b]_t! [a]_t!)$ and $\begin{bmatrix} a+b \\ b-1 \end{bmatrix}_t = [a+b]_t! / ([b-1]_t! [a+1]_t!)$,

$$\begin{bmatrix} a+b \\ b \end{bmatrix}_t - \begin{bmatrix} a+b \\ b-1 \end{bmatrix}_t = \frac{[a+b]_t!}{[b]_t! [a+1]_t!} ([a+1]_t - [b]_t).$$

The bracket telescopes: $[a+1]_t - [b]_t = t^b + t^{b+1} + \dots + t^a = t^b [a-b+1]_t$. Substituting matches the fake-degree expression above. $\square$

3 The telescoping identity

Set $c_j := \begin{bmatrix} 2k \\ j \end{bmatrix}_t$ (so $c_j \in \mathbb{Z}[t]$) and $c_{-1} := 0$. Define the alternating partial sum

$$S_j(t) := \sum_{i=0}^j (-1)^i c_i = \sum_{i=0}^j (-1)^i \begin{bmatrix} 2k \\ i \end{bmatrix}_t.$$

Proposition 4. For every $k \geq 1$,

$$m_{(2)}(t) = S_j(t), \quad m_{(1,1)}(t) = -S_{j'}(t),$$

where

$$(j, j') = \begin{cases} (k, k-1) & k \text{ even,} \\ (k-1, k) & k \text{ odd.} \end{cases}$$

Consequently

$$m_{(2)}(t) + m_{(1,1)}(t) = \begin{bmatrix} 2k \\ k \end{bmatrix}_t.$$

Proof. By Lemma 3, $\tilde{f}_{(2k-2i, 2i)}(t) = c_{2i} - c_{2i-1}$, so

$$m_{(2)}(t) = \sum_{i=0}^K (c_{2i} - c_{2i-1}) = c_0 - c_1 + c_2 - c_3 + \dots + (-1)^{2K} c_{2K} = \sum_{i=0}^{2K} (-1)^i c_i = S_{2K}(t),$$

with $K = \lfloor k/2 \rfloor$; and $2K = k$ if k even, $2K = k-1$ if k odd.

Similarly $\tilde{f}_{(2k-2i-1, 2i+1)}(t) = c_{2i+1} - c_{2i}$, so

$$m_{(1,1)}(t) = \sum_{i=0}^{K'} (c_{2i+1} - c_{2i}) = - \sum_{i=0}^{2K'+1} (-1)^i c_i = -S_{2K'+1}(t),$$

with $K' = \lfloor (k-1)/2 \rfloor$; and $2K' + 1 = k$ if k odd, $k-1$ if k even.

For the final identity, note that $S_k(t) - S_{k-1}(t) = (-1)^k c_k$. If k is even, $m_{(2)} + m_{(1,1)} = S_k - S_{k-1} = c_k$; if k is odd, $m_{(2)} + m_{(1,1)} = S_{k-1} - S_k = -(-1)^k c_k = c_k$. Either way the sum equals $c_k = \left[\begin{smallmatrix} 2k \\ k \end{smallmatrix} \right]_t$. (Equivalently: this is $h_2[h_k] + e_2[h_k] = h_k^2$ on the level of fake degrees, using the Pieri expansion $h_k^2 = \sum_{j=0}^k s_{(k+j, k-j)}$ and the MacMahon identity $\sum_j \tilde{f}_{(k+j, k-j)}(t) = \left(\begin{smallmatrix} 2k \\ k \end{smallmatrix} \right)_t$.) $\square$

Remark. The identity $\sum_{j=0}^k \tilde{f}_{(k+j, k-j)}(t) = \left(\begin{smallmatrix} 2k \\ k \end{smallmatrix} \right)_t$ is the classical MacMahon formula for the Hilbert series of the parabolic coinvariant $R_{(k,k)} = R_{2k}^{S_k \times S_k}$; it is also Theorem D's total Hilbert series identity at $r = 2$.

4 The q -Lucas evaluation

Let $\omega = e^{2\pi i/3}$ so $[3]_t = 1 + t + t^2 = \Phi_3(t)$ has roots $\{\omega, \bar{\omega}\}$. Since $[3]_t = \Phi_3(t)$ is irreducible over $\mathbb{Q}[t]$, for $f \in \mathbb{Q}[t]$:

$$[3]_t \mid f(t) \iff f(\omega) = 0.$$

We use the following “correct” form of the q -Lucas theorem.

Lemma 5 (q -Lucas at ζ_d , Désarménien 1982). *For any positive integer d , any primitive d -th root of unity ζ , and any nonnegative integers $n \geq k$,*

$$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_t \Big|_{t=\zeta} = \left(\left[\begin{smallmatrix} n/d \\ k/d \end{smallmatrix} \right] \right) \cdot \left[\begin{smallmatrix} n \bmod d \\ k \bmod d \end{smallmatrix} \right]_t \Big|_{t=\zeta},$$

where the first factor is an ordinary binomial coefficient and the second is a q -binomial evaluated at ζ .

Specialising to $d = 3$ and $\zeta = \omega$, the possible inner-factor values are

$$\left[\begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right]_{t\omega} = 1, \quad \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix} \right]_{t\omega} = \left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix} \right]_{t\omega} = 1, \quad \left[\begin{smallmatrix} 2 \\ 0 \end{smallmatrix} \right]_{t\omega} = \left[\begin{smallmatrix} 2 \\ 2 \end{smallmatrix} \right]_{t\omega} = 1, \quad \left[\begin{smallmatrix} 2 \\ 1 \end{smallmatrix} \right]_{t\omega} = 1 + \omega = -\omega^2,$$

and $\left[\begin{smallmatrix} r \\ s \end{smallmatrix} \right]_{t\omega} = 0$ whenever $s > r$ within $\{0, 1, 2\}$.

Denote $B_r := (-1)^r \left[\begin{smallmatrix} 2 \bmod 3 \\ r \end{smallmatrix} \right]_t \Big|_{t\omega}$, the sign-weighted inner-factor values used when $2k \equiv n \bmod 3$. There are three regimes depending on $n \bmod 3 \in \{0, 1, 2\}$:

- ($n \equiv 0$): $B_0 = 1$, $B_1 = B_2 = 0$.
- ($n \equiv 1$): $B_0 = 1$, $B_1 = -1$, $B_2 = 0$. **The full-triple sum** $B_0 + B_1 + B_2 = 0$.
- ($n \equiv 2$): $B_0 = 1$, $B_1 = -(-\omega^2) = \omega^2$, $B_2 = 1$; full-triple sum $= 2 + \omega^2 = 1 - \omega \neq 0$.

Since $n = 2k$: $k \equiv 0 \pmod{3} \Leftrightarrow n \equiv 0$; $k \equiv 1 \pmod{3} \Leftrightarrow n \equiv 2$; $k \equiv 2 \pmod{3} \Leftrightarrow n \equiv 1$.

5 Proof of Theorem 2

By Proposition 4, both $m_{(2)}(t)$ and $m_{(1,1)}(t)$ are, up to sign, of the form $S_j(t)$ with $j \in \{k-1, k\}$. It suffices to show

- (a) $\left[\begin{smallmatrix} 2k \\ k \end{smallmatrix} \right]_t |_\omega = 0 \iff k \equiv 2 \pmod{3}$;
- (b) at $k \equiv 2 \pmod{3}$, $m_{(2)}(\omega) = 0$;
- (c) at $k \equiv 0 \pmod{3}$ or $k \equiv 1 \pmod{3}$, $m_{(2)}(\omega) \neq 0$.

Combining (a) with (b) gives $m_{(2)}(\omega) = m_{(1,1)}(\omega) = 0$ (using the sum identity), so both $[3]_t$ -divide; that's the $\Leftarrow$ direction of Theorem 2. Item (c) gives the $\Rightarrow$ direction (at least $m_{(2)}$ fails).

Proof of (a). By Lemma 5, $\binom{2k}{k}_\omega = \binom{\lfloor 2k/3 \rfloor}{\lfloor k/3 \rfloor} \cdot \left[\begin{smallmatrix} 2k \bmod 3 \\ k \bmod 3 \end{smallmatrix} \right]_t |_\omega$. The inner factor is $\begin{bmatrix} 0 \\ 0 \end{bmatrix}_t |_\omega = 1$ ($k \equiv 0$), $\begin{bmatrix} 2 \\ 1 \end{bmatrix}_t |_\omega = -\omega^2 \neq 0$ ($k \equiv 1$), or $\begin{bmatrix} 1 \\ 2 \end{bmatrix}_t |_\omega = 0$ ($k \equiv 2$). Hence the product is zero iff $k \equiv 2 \pmod{3}$.

Proof of (b). Write $k = 3s + 2$, $n = 2k = 6s + 4 \equiv 1 \pmod{3}$; so $\lfloor n/3 \rfloor = 2s + 1$ and $n \bmod 3 = 1$. By Lemma 5,

$$c_i |_\omega = \left[\begin{smallmatrix} 2k \\ i \end{smallmatrix} \right]_t |_\omega = \binom{2s+1}{\lfloor i/3 \rfloor} \cdot \left[\begin{smallmatrix} 1 \\ i \bmod 3 \end{smallmatrix} \right]_t |_\omega.$$

The inner factor is 1 if $i \bmod 3 \in \{0, 1\}$ and 0 if $i \bmod 3 = 2$. Splitting $i = 3q + r$,

$$S_j(\omega) = \sum_{q: 3q \leq j} (-1)^q \binom{2s+1}{q} \cdot \underbrace{[B_0 \cdot \mathbb{K}[3q \leq j] + B_1 \cdot \mathbb{K}[3q+1 \leq j]]}_{=: E_q},$$

where $B_0 = 1$, $B_1 = -1$ (the B_2 -term is absent since the inner factor $\begin{bmatrix} 1 \\ 2 \end{bmatrix}_t |_\omega = 0$).

For any q with $3q + 1 \leq j$, $E_q = B_0 + B_1 = 0$; only the “last” q (where the constraint tightens) can contribute.

For $\rho := j - 3\lfloor j/3 \rfloor \in \{0, 1, 2\}$:

- $\rho = 0$: at $q = j/3$, $3q = j$ but $3q + 1 = j + 1 > j$; $E_q = B_0 = 1$; contribution $(-1)^{j/3} \binom{2s+1}{j/3}$.
- $\rho = 1$: at $q = (j-1)/3$, both $3q = j-1 \leq j$ and $3q+1 = j \leq j$; $E_q = B_0 + B_1 = 0$. *No contribution.*
- $\rho = 2$: at $q = (j-2)/3$, both $3q+1 \leq j$; $E_q = 0$. *No contribution.*

Now split by parity of $k = 3s + 2$:

- k even $\implies s$ even, so $k = 6t + 2$; $j = k = 6t + 2$; $\rho = 6t + 2 \bmod 3 = 2$. Contribution zero: $S_j(\omega) = 0$.
- k odd $\implies s$ odd, so $k = 6t + 5$; $j = k - 1 = 6t + 4$; $\rho = (6t + 4) \bmod 3 = 1$. Contribution zero: $S_j(\omega) = 0$.

Either way, $m_{(2)}(\omega) = S_j(\omega) = 0$. □(b)

Proof of (c). Two subcases.

Case $k \equiv 0 \pmod{3}$. Write $k = 3s$, $n = 2k = 6s \equiv 0 \pmod{3}$; $\lfloor n/3 \rfloor = 2s$. Inner factor $\begin{bmatrix} 0 \\ r \end{bmatrix}_t |_\omega$ is 1 if $r = 0$ else 0. Hence $c_i |_\omega = \binom{2s}{i/3}$ if $3 \mid i$, and 0 otherwise. Thus

$$S_j(\omega) = \sum_{q=0}^{\lfloor j/3 \rfloor} (-1)^{3q} \binom{2s}{q} = \sum_{q=0}^Q (-1)^q \binom{2s}{q},$$

with $Q = \lfloor j/3 \rfloor$. The classical alternating identity $\sum_{q=0}^Q (-1)^q \binom{N}{q} = (-1)^Q \binom{N-1}{Q}$ (for $Q < N$) gives:

- k even $= 6u$: $j = k = 6u$; $Q = 2u$; $2s = 4u$; $S_j(\omega) = \binom{4u-1}{2u}$. Positive for $u \geq 1$.
- k odd $= 6u + 3$: $j = k - 1 = 6u + 2$; $Q = 2u$; $2s = 4u + 2$; $S_j(\omega) = \binom{4u+1}{2u}$. Positive for $u \geq 0$.

Both are positive integers, so nonzero, so $m_{(2)}(\omega) \neq 0$.

Case $k \equiv 1 \pmod{3}$. Write $k = 3s + 1$, $n = 2k = 6s + 2 \equiv 2 \pmod{3}$; $\lfloor n/3 \rfloor = 2s$. Inner factor: $B_0 = 1$, $B_1 = \omega^2$ ($= -(-\omega^2)$), $B_2 = 1$. Full-triple sum: $B_0 + B_1 + B_2 = 2 + \omega^2 = 1 - \omega$ (using $1 + \omega + \omega^2 = 0$).

Split $i = 3q + r$: $c_i |_\omega = \binom{2s}{q} \cdot \begin{bmatrix} 2 \\ r \end{bmatrix}_t |_\omega$. For q with $3q + 2 \leq j$ (all three r 's in range): $E_q^{\text{full}} = 1 - \omega$. For the boundary q :

- $\rho = 0$: $E_Q = 1$;
- $\rho = 1$: $E_Q = 1 + \omega^2 = -\omega$;
- $\rho = 2$: full ($E_Q = 1 - \omega$).

Split by parity of $k = 3s + 1$:

- k even $= 6u + 4$ (so $s = 2u + 1$, $2s = 4u + 2$): $j = k = 6u + 4$; $Q = 2u + 1$; $\rho = 1$.

$$\begin{aligned} S_j(\omega) &= (1 - \omega) \sum_{q=0}^{2u} (-1)^q \binom{4u+2}{q} + (-1)^{2u+1} \binom{4u+2}{2u+1} \cdot (-\omega) \\ &= (1 - \omega) \binom{4u+1}{2u} + \omega \binom{4u+2}{2u+1}. \end{aligned}$$

Now $\binom{4u+2}{2u+1} = 2 \binom{4u+1}{2u}$ (Pascal's rule $\binom{N+1}{K+1} = \binom{N}{K} + \binom{N}{K+1}$ with $N = 4u + 1$, $K = 2u$ and symmetry $\binom{4u+1}{2u} = \binom{4u+1}{2u+1}$), so

$$S_j(\omega) = \binom{4u+1}{2u} [(1 - \omega) + 2\omega] = \binom{4u+1}{2u} (1 + \omega) = -\omega^2 \binom{4u+1}{2u}.$$

For $u \geq 0$: $\binom{4u+1}{2u} \geq 1$, so $|S_j(\omega)| = \binom{4u+1}{2u} > 0$.

- k odd $= 6u + 1$ (so $s = 2u$, $2s = 4u$): $j = k - 1 = 6u$; $Q = 2u$; $\rho = 0$.

$$\begin{aligned} S_j(\omega) &= (1 - \omega) \sum_{q=0}^{2u-1} (-1)^q \binom{4u}{q} + (-1)^{2u} \binom{4u}{2u} \cdot 1 \\ &= -(1 - \omega) \binom{4u-1}{2u-1} + \binom{4u}{2u}. \end{aligned}$$

Using $\binom{4u}{2u} = 2 \binom{4u-1}{2u-1}$ (Pascal / symmetry as above),

$$S_j(\omega) = \binom{4u-1}{2u-1} [-(1 - \omega) + 2] = \binom{4u-1}{2u-1} (1 + \omega) = -\omega^2 \binom{4u-1}{2u-1}.$$

For $u \geq 1$: nonzero. For $u = 0$ (i.e., $k = 1$): $S_0(\omega) = c_0 = 1 \neq 0$ directly.

So $S_j(\omega) = -\omega^2 \cdot (\text{positive integer}) \neq 0$ in all cases.

In both subcases (c), $m_{(2)}(\omega) \neq 0$; hence $[3]_t \nmid m_{(2)}(t)$. Theorem 2 is proved. $\square$

6 Computational verification

The following are verified by SymPy in `~/projects/probes/2026-07-31-r2-proof-verification/verify.py`:

1. Reduction identity of Proposition 4: $m_{(2)}(t)$ and $m_{(1,1)}(t)$ computed directly via Theorem 1's isotypic sum vs. via $\pm S_j(t)$ agree as polynomials in $\mathbb{Z}[t]$ for all $k \in \{1, \dots, 8\}$ (zero mismatches).
2. Divisibility pattern: for $k \in \{1, \dots, 12\}$, $[3]_t \mid m_\pi(t)$ for both $\pi \in \{(2), (1,1)\}$ iff $k \equiv 2 \pmod{3}$ (verified by polynomial long division; zero mismatches).
3. The q -Lucas evaluation of $\left[\begin{smallmatrix} 2k \\ i \end{smallmatrix} \right]_t |_\omega$ agrees symbolically with Lemma 5 for $k \in \{2, \dots, 6\}$ and all $i \in \{0, \dots, 2k\}$ (all direct computations = the ordinary-binomial-times-inner-factor product).
4. The closed-form values of $S_j(\omega)$ from the proof of (c) agree numerically with the direct SymPy evaluation for $k \in \{1, \dots, 12\}$, up to the phase factor $-\omega^2$ (which has $|\omega^2| = 1$).

7 Discussion and outlook

The proof structure is:

$$\text{isotypic sums} \xrightarrow[\text{Lemma 2}]{\text{two-row fake deg}} \text{telescope of } q\text{-binomials} \xrightarrow[\text{Lemma 4}]{q\text{-Lucas}} \text{finite } \omega\text{-arithmetic}.$$

The vanishing in case (b) is not the vanishing of any individual $\tilde{f}_\lambda(\omega)$ (indeed we know from the probe that the individual fake degrees typically do *not* divide by $[3]_t$); it is a coherent cancellation induced by the alternating structure of the telescope and by the fact that the “inner” full-triple q -binomial sum $\left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix} \right]_t |_\omega - \left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix} \right]_t |_\omega + 0 = 0$ vanishes precisely when $n \equiv 1 \pmod{3}$ (i.e., $k \equiv 2$).

The pattern of the survivor in case (c) is striking: the residual value is always $-\omega^2 \cdot \binom{2s' \pm 1}{s'}$ for a central binomial coefficient. Both cases $k \equiv 0$ and $k \equiv 1$ produce central-binomial magnitudes with real vs. complex- ζ_3 -phase, and the magnitudes are all positive integers.

Remark (Generalisation to $r \geq 3$). The full conjecture ([3]) states $[2r-1]_t \mid m_\pi(t)$ for every $\pi \vdash r$ iff $k \bmod (2r-1) \in \{2, \dots, 2r-2\}$. The $r = 2$ case above is the smallest instance. For $r \geq 3$ the telescoping argument does not generalise directly; one likely route is Springer's theorem on regular elements applied to the wreath $S_k \wr S_r$, or the wreath-plethystic formula of Macdonald Ch I App B combined with the fake-degree formula at ζ_d (Désarménien).

References

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- [2] C. Muse, *The r -wreath extension of Theorem D*, note (2026-07-30).
- [3] `~/state/PROVE.md.completed-20260731`, period- $(2r-1)$ divisibility conjecture.